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Combining dynamic life cycle assessment and net ecosystem exchange through the framework of biobased materials and products-life cycle assessment (BBM-LCA): application to polylactic acid

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Received November 29 2024; Revised March 15 2025; Accepted March 18 2025;
View online 9 May 2025 at Wiley Online Library (wileyonlinelibrary.com);
DOI: 10.1002/bbb.2769; *Biofuels, Bioprod. Bioref.* 19:1075–1087 (2025)

Abstract: Biobased products can achieve carbon negativity by storing biogenic carbon in the technosphere through circular loops. Quantifying these benefits requires quantitative assessment tools such as life cycle assessment (LCA). However, the LCA of biobased materials is not straightforward due to the complexity of addressing agroecosystem carbon dynamics and the timing of emissions. The aim of this research is to apply the novel framework of biobased materials and products-life cycle assessment (BBM-LCA) to the example of polylactic acid (PLA) to show how it can address biogenic carbon and timing issues of emissions by combining net ecosystem exchange (NEE) and dynamic LCA. BBM-LCA adapts two aspects of conventional LCA to estimate the climate impact of biobased products more accurately. First, concerning the absorption of atmospheric carbon during cultivation, BBM-LCA uses NEE to account for all biogenic emissions occurring within the agricultural ecosystem. This means that it is not limited to the absorption of carbon by agricultural crops. Second, concerning the calculation of climate impacts, BBM-LCA adopts dynamic LCA as a time-dependent approach, to generate a time-sensitive global warming potential (GWP). The example of PLA proves that BBM-LCA is an effective instrument for calculating the climate impact of biobased products due to the implementation of a holistic lifecycle approach and a dynamic impact calculation method. BBM-LCA accounts for the carbon sequestration benefits of recycling, recognizing its actual impact over time in multiple lifecycles. This feature makes BBM-LCA preferable over conventional LCA, which struggles to track greenhouse gas (GHG) emissions at different points over multiple years across the multiple lifecycles of recycled products. © 2025 The

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Key words: environmental sustainability; dynamic LCA; net ecosystem exchange; biogenic carbon; polylactic acid; global warming potential

Introduction

Biobased products are composed of 'biogenic' carbon, which is carbon derived from plant or animal sources – excluding fossil carbon and peat – sequestered from the atmosphere during biomass growth.[1] Biobased products have the potential to be not only carbon neutral but also carbon negative through the storage of biogenic carbon in the technosphere, particularly when long-term storage strategies are pursued (e.g., reuse in multi-cycle circular loops). To quantify such advantages, a dynamic environmental life cycle assessment (LCA) must be used as static approaches fail to address the time dynamics of biogenic carbon.

The first static approach, which is referred to as the '0/0 approach' or 'carbon neutral approach', is based on the assumption that the release of CO_2 from a biobased product at the end of its life is balanced by an equivalent uptake of CO_2 during biomass growth (i.e. biogenic CO_2 uptake (0) and release (0)). In the second static approach, which is referred to as the '−1/+1' approach, the uptake of biogenic CO_2 during the biomass growth is reported as a negative emission (−1) and the release of biogenic CO_2 at the end of life is reported as a positive emission (+1).

The main shortcoming of static approaches is their inability to account for temporal dynamics, which can be problematic when assessing the impact of biobased products. For long-lived biobased products, where greenhouse gas (GHG) emissions occur at different times, the LCA timeframe may not align with the global warming potential (GWP) time horizon. For example, with a 100 year time horizon, emissions from a product with a 25 year lifespan would be assessed from year 25 to year 125. Assuming that carbon neutrality equals climate neutrality also neglects the time dynamics of biogenic carbon emissions and sequestration. Given the time lag between biomass removal and regrowth, slow-growing agricultural crops cannot be considered carbon neutral. Conversely, fast-growing agricultural crops can remove carbon from the atmosphere rapidly. A carbon neutrality assumption also ignores the benefit that biobased products store carbon during the product's lifecycle. Overall, the static approaches will not allow us to (i) differentiate between slow-growing agricultural crops and fast-growing ones or (ii) give credit to the long-lived biobased products.

To better capture the impact of time, dynamic approaches have been developed. Levasseur *et al.*[2] developed dynamic LCA, which incorporates time-sensitivity into LCAs by using a temporally resolved inventory of lifecycle greenhouse gas emissions. For example, if an amount X of CO_2 is emitted in the year of product manufacturing, and an amount Y of CO_2 is emitted 10 years down the line for product maintenance, the timing of these emissions must be considered alongside their respective amounts.

In the context of biobased materials and products, dynamic LCA implies that all emissions, ranging from feedstock cultivation down to product disposal, must be precisely and time-sensitively inventoried. For an LCA practitioner, precisely accounting for GHG emissions occurring during the feedstock cultivation phase can lead to potential challenges due to the complex transfer of carbon between the agroecosystem and the atmosphere.[3] Agroecosystem-atmosphere GHG exchanges are not simply limited to the absorption of CO_2 from growing the agricultural crop but also include the cycling of carbon within the soil, a complex phenomenon that depends on numerous factors (e.g., the initial stock of carbon present under the land, anthropogenic agricultural practices, and climate conditions).[4–6] The importance of quantifying GHG emissions from soil has been highlighted in the literature.[7–9] There is a simple notion of net ecosystem exchange (NEE), which covers all ecosystem carbon exchanges and presents the exchange of CO_2 between the ecosystem and the atmosphere directly.[10] When measurements of NEE are not possible, they can be estimated through ecosystem simulation models,[11] where soil carbon models play an important role. Soil carbon models require relatively little input data and are accessible to life cycle analysts.[12] They also have an advantage over static approaches in that they consider the dynamics of soil carbon and thereby better describe land-based CO_2 emissions.[4,13]

Shen *et al.*[14] have recently applied dynamic approaches to assess the sustainability of biobased products, taking into account the impacts of temporary carbon storage and dynamically accounting for soil carbon emissions. Shen *et al.*[14] coined the term 'carbon storage in arable land and anthropogenic products' (CSAAP), where a full ecosystem and product framework was adopted to discuss the climate benefits of biobased materials and their renewable feedstocks. In their

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holistic approach, biogenic carbon is divided among four pools: the air, the soil, the plants, and the technosphere. The complex carbon flows within the soil are simulated using a carbon model, and the temporal impacts of carbon storage within products are evaluated using a dynamic calculation method. However, the emphasis of the study by Shen *et al.*[14] was on finding an ideal feedstock for climate mitigation purposes. A standardized LCA methodology was thus not adopted for the sustainability assessment of biobased materials, making the CSAAP approach difficult to apply directly to LCA studies.

More recently, Kouamé[15] developed a new framework of 'biobased materials and products-life cycle assessment' (BBM-LCA), which can be directly applicable to the field of LCA. The framework is designed to combine both a holistic ecosystems approach, and a temporally dynamic calculation methodology. This framework was intended to adopt the principles of life cycle thinking and to be incorporated directly into the ISO-14040[16] standardized LCA for biobased materials and products. Kouamé[15] applied BBM-LCA to a fictitious case study in order to illustrate how BBM-LCA works for only limited end-of-life (EoL) options, which does not include the composting, digestion and chemical recycling scenarios. Recently, BBM-LCA was applied to the cradle-to-gate case of polylactic acid (PLA), by Ghannadzadeh and van der Meer,[17] where BBM-LCA could effectively address the time dependency of each unit of the biobased products value chain.

The scope of cradle-to-grave including the recycling scenarios has its own complexities, which have been pointed out briefly by the application of BBM-LCA on recycling scenarios in El Kadi *et al.*,[18] where the details such as material loss in chemical recycling were neglected. To address the complexities associated with the common end-of-life scenarios for a typical biobased plastic in depth, this study presents the BBM-LCA framework based on El Kadi *et al.*[18] in a more detailed analysis – for example, the material loss in chemical recycling is included. This brings another dimension to this research as the time dynamics of recycling scenarios could not be addressed in the static LCA studies. Overall, this study will not only present the benefits of coupling of NEE with dynamic LCA but also the added values of the BBM-LCA for the modeling of recycling scenarios.

The paper starts by introducing the BBM-LCA framework's methodology, where the NEE and dynamic LCA are presented followed by an example of PLA. Carbon exchange through the soil and sugar beets is calculated for the specific case of PLA through the Andriulo–Mary–Guérif (AMG) model[22]. Finally, the dynamic inventory is translated to instantaneous and cumulative impacts through dynamic LCA, which enables us to analyze the entire value chain using a dynamic full-ecosystem approach.

Materials and methods

The BBM-LCA framework

Overview

The structure and working principles of the BBM-LCA framework are described based on Kouamé.[15] BBM-LCA is an LCA-based methodology for assessing the climate impacts of biobased materials. Its aim is to operate within the framework of standardized LCA and carbon footprint calculations while presenting additional precision for the specific case of biobased materials and products. The BBM-LCA approach takes into account ISO 14040 standards regarding the practice of LCA but adds an ecosystem-oriented and dynamic approach to assess biogenic carbon emissions.

BBM-LCA, depicted in Fig. 1, respects the four phases of an LCA study as detailed in the ISO 14040 standard. First, the goal and scope of the study need to be established. Certain elements of this initial phase are specified by the BBM-LCA methodology, whereas others are open to the discretion of the life cycle analyst. For instance, BBM-LCA is intrinsically limited to a single impact category: the effects on climate change. It requires that a cradle-to-grave approach is adopted. With respect to the goal of the study, nevertheless, different scenarios are possible, such as policymaking or sustainable product development. Finally, an important element of the goal and scope is the definition of a functional unit and the choice of a corresponding reference product that meets the requirements of the functional unit. Although the functional unit is open to the choice of the analyst, and can be based on the requirements of the study, the corresponding reference products must be biobased materials or products.

Following the phase of goal and scope definition, the next step is the life cycle inventory (LCI). During this phase, all inputs and outputs for the reference products are collected and compiled. In the case of BBM-LCA, the procedure for carrying out the LCI phase includes an additional layer of detail: the LCI should be carried out in a time-sensitive manner, where inputs and outputs are associated with a time signature describing the moment at which they happen. For example, if a yearly time step is considered, the LCI will involve compiling inputs and outputs year by year. The resulting inventory is therefore structured as a list of yearly inputs and outputs, as opposed to a single value aggregated over the lifecycle, which is usual in conventional LCAs. The approach to LCIs proposed by BBM-LCA also differs from conventional LCIs in how it accounts for emissions of biogenic carbon. Such emissions are very important in the context of biobased materials and products: they

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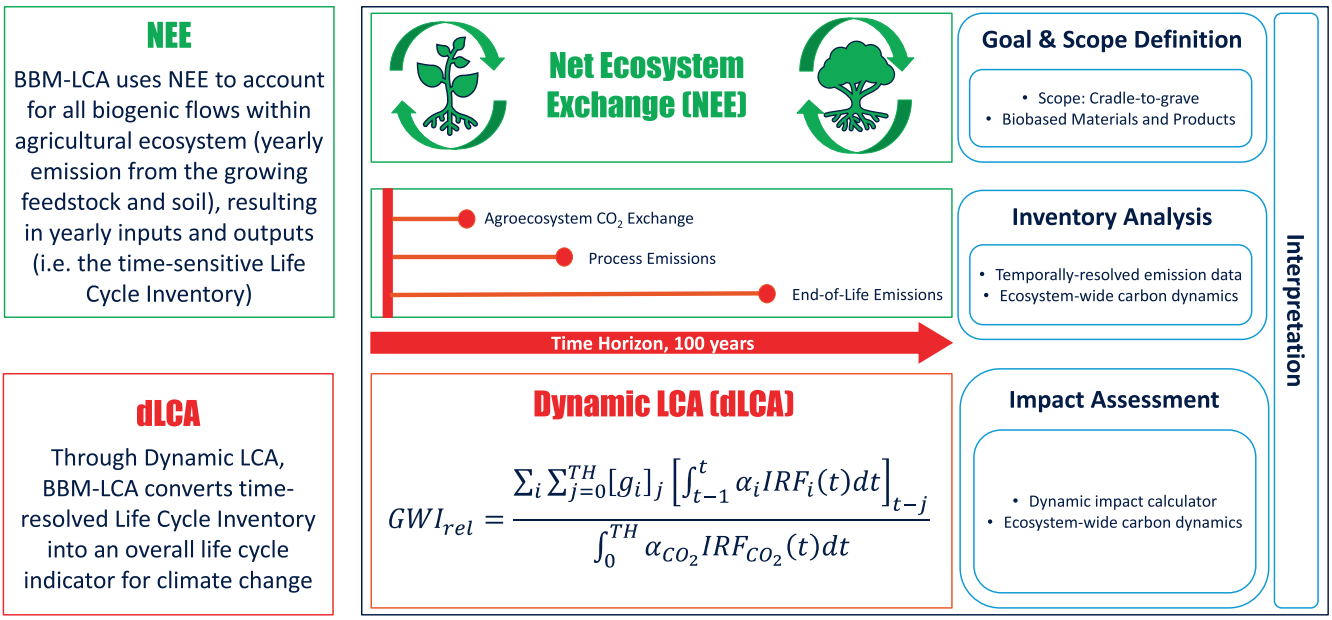


Figure 1. The new framework of biobased materials and products-life cycle assessment (BBM-LCA).

happen as negative emissions (i.e. absorptions) during the growth of the biological feedstock and as positive emissions during the incineration of the final disposed product. BBM-LCA proposes an ecosystem approach to deal with the absorption of carbon through the growth of biobased feedstock rather than simply taking into account the removal of atmospheric CO_2 by the growing agricultural crop. It also considers potential ecosystem ramifications resulting from the harvesting activity, which is notably emissions and absorptions of carbon by the soil. In order to do this precisely, the proposed methodology suggests the use of NEE and recommends using ecosystem simulation models to estimate NEE.

In the third phase, the life cycle impact assessment (LCIA), inputs and outputs are translated from amounts of a given substance into potential impacts on the environment. For instance, taking climate change as the environmental impact observed, the LCIA phase will convert amounts of GHG into warming effects on the planet. BBM-LCA operates as a single-category impact indicator, generating indications for impact on climate change in the form of a global warming potential analogous to the Intergovernmental Panel on Climate Change's (IPCC's) conventional GWP.[19] Through the application of a time-sensitive methodology, BBM-LCA translates time-resolved inventories of GHG emissions into an overall life cycle indicator for climate change. The mathematical approach used to achieve this was developed by Levasseur *et al.*[2] It is important to note that the current version of BBM-LCA presented here only takes into account biobased materials' effect on climate change whereas additional impact categories can be considered in theory.

The final stage of LCA, the life cycle interpretation stage, remains unchanged in the proposed methodology.

Overall, BBM-LCA mainly adapts two aspects of conventional LCA studies in order to more accurately estimate the climate impact of biobased plastics. First, regarding the absorption of atmospheric carbon during the feedstock cultivation, BBM-LCA uses the concept of NEE to account for all biogenic emissions occurring within the agricultural ecosystem, which is not only the absorption of carbon by the growing agricultural crop. Second, concerning the calculation of climate impacts, BBM-LCA adopts a time --dependent approach of dynamic LCA to generate a time-sensitive product GWP.

Modeling of agroecosystem carbon dynamics in LCA

Ecosystem CO_2 emissions and the concept of NEE

During photosynthesis, carbon present in the atmosphere in the form of carbon dioxide is absorbed by growing plants in the form of sugars. During respiration, carbon dioxide is emitted back into the atmosphere as sugars are consumed for their energy content. Two forms of respiration occur within an ecosystem. On the one hand, autotrophic respiration is

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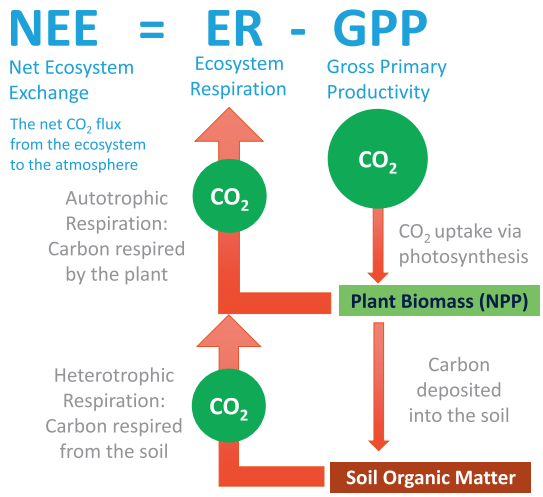


Figure 2. Net ecosystem exchange (NEE).

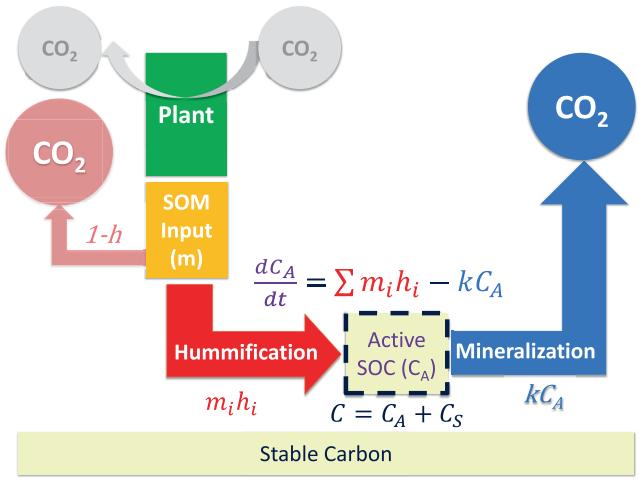


Figure 3. Modeling the carbon exchange within the active pool of soil carbon.

carried out by plants as they consume part of the energy that they produce through photosynthetic production of sugars. The remaining part is contained within their biomass and consumed by the ecosystem's non-plant organisms in the process referred to as heterotrophic respiration.[20] Not all the carbon is re-emitted as some plant material is held permanently within the ecosystem's soil. As Fig. 2 illustrates, the total amount of carbon stored by an ecosystem is the difference between the carbon it absorbs from the atmosphere and the carbon it respires.

Gross primary productivity (GPP) is relatively easy to estimate on the basis of crop yield. Net primary productivity (NPP) has also been calculated by Bolinder *et al.*[21] However, soil respiration occurs through a complex process dependent on the interaction between various climatic and soil characteristics. Thus, BBM-LCA makes provision for the use of a soil carbon model to simulate soil respiration.

The AMG model

The AMG model[22] divides the soil carbon into two pools: (i) active carbon, which is released back into the atmosphere after it is decomposed by microorganisms, and (ii) stable carbon, which is stored in the soil for a long time. Soil organic carbon (SOC) refers to the carbon stored in the soil. Its primary source is soil organic matter (SOM) as shown in Fig. 3. The rate, h, at which SOM turns into SOC is called the hummification rate. As not all the SOM is decaying, not all of it will be converted into humus, meaning that SOM can still release carbon without being part of the active pool. This faction of carbon is described by 1-h. The remaining carbon is preserved in the active pool, where it is emitted over a certain period depending on k, the rate of mineralization, which is breakdown of organic matter into inorganic matter. The variable k denotes the rate at which the SOC of the active pool is released. The balancing dynamics of hummification and mineralization describe the carbon transfer in the active pool from the soil as well as the plant.

Dynamic LCA

The concept and governing equations of dynamic LCA are presented in this section, based on Levasseur *et al.*[2] The current practice is to use GWP, adopted by the IPCC as a characterization factor for global warming impact assessment. The GWP expresses the cumulative radiative forcing value caused by emission of a unit mass of a given GHG over a defined time horizon, relative to the equivalent value for CO_2 (Eqn 1). TH is the chosen time horizon, α is the instantaneous radiative forcing per unit mass increase in the atmosphere.

$$GWP_i^{TH} = \frac{\int_0^{TH} \alpha_i \mathrm{IRF}_i(t)dt}{\int_0^{TH} \alpha_{\mathrm{CO}_2} \mathrm{IRF}_{\mathrm{CO}_2}(t)dt} \quad (1)$$

Computing the time-dependent impact on global warming (GWI) needs a dynamic inventory and yearly dynamic characterization factors (DCFs). To obtain the yearly DCFs, the time scale was divided into 1 year time steps and the integration boundaries were set for every time step (Eqn 2). As Fig. 4 shows, the dynamic inventory is obtained by dividing the lifecycle into 1-year time steps and by adding the different emissions of each GHG occurring at every time step.

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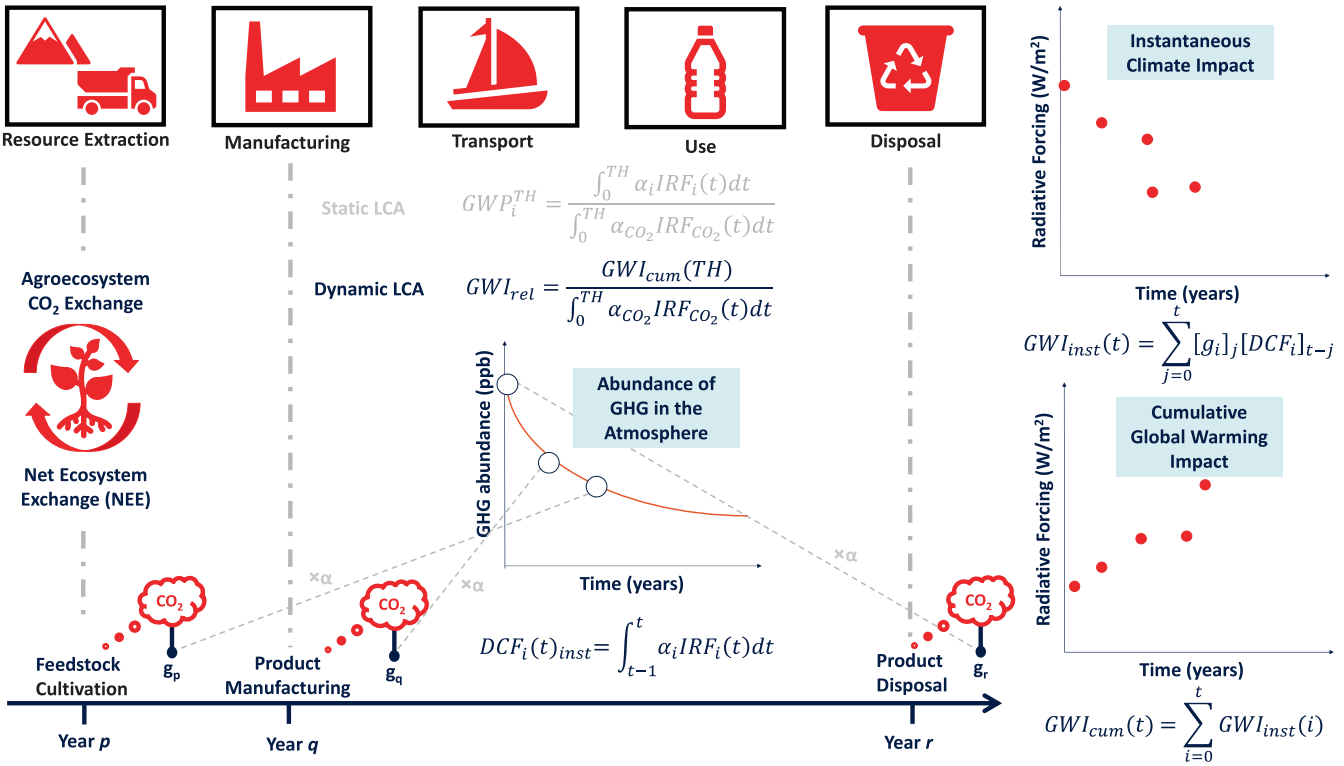


Figure 4. Translating dynamic LCI to GWP through dynamic LCA.

$$\mathrm{DCF}_i(t)_{\mathrm{inst}} = \int_{t-1}^{t} \alpha_i \mathrm{IRF}_i(t)dt \quad (2)$$

GWI needs to represent the cumulative radiative forcing since the emission. GWI_{inst} occurring at t is obtained by summing the radiative forcing occurring at t caused by each previous lifecycle emission. This is determined by multiplying each emission by DCF calculated for the period elapsed between the emission and t (Eqn 3).

$$GWI_{inst}(t) = \sum_{j=0}^{t} [g_i]_j [DCF_i]_{t-j} \quad (3)$$

$GWI_{\mathrm{cum}}(t)$, is the sum of the $\mathrm{GWI}_{\mathrm{inst}}$ calculated for all the previous years. $\mathrm{GWI}_{\mathrm{cum}}(t)$ expresses the total amount of increased radiative forcing caused by all emissions over a given time period (Eqn 4).

$$\mathrm{GWI}_{\mathrm{cum}}(\mathrm{t}) = \sum_{\mathrm{i}=0}^{\mathrm{t}} \mathrm{GWI}_{\mathrm{inst}}(\mathrm{i}) \quad (4)$$

To compare $GWI_{cum}(t)$ with the results of the traditional LCA, a time horizon is chosen (TH). $GWI_{cum}(TH)$ is divided by the cumulative radiative forcing of a 1 kg CO_2 emission at $t = 0$ (Eqn 5).

$$GWI_{rel} = \frac{GWI_{cum}(TH)}{\int_0^{TH} \alpha_{CO_2} IRF_{CO_2}(t)dt} \quad (5)$$

Application of BBM-LCA: an example of a rug made from PLA

In this section, BBM-LCA is applied to a fictitious example, where the goal is to calculate the carbon footprint of a rug made from PLA under six EoL scenarios: chemical recycling-depolymerization[33], chemical recycling-hydrolysis[32], mechanical recycling[32], composting[32], anaerobic digestion[33], incineration[23]. This means that the scope of this example is cradle-to-grave. The base case EoL treatment is considered to be incineration with energy recovery. The functional unit of this example is 'providing the floor cover for 100 years', which means that the rug with an average lifespan of 25 years needs to be replaced four times. The reference flow is taken to be 1 kg of PLA. The static LCI including the data related to PLA production and EoL options is based on literature data

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Table 1. Parameters used in the AMG model and the Bolinder *et al.*[21] model.

Category	Parameter	Description	References
Parameters of AMG model	T = 10.7 °C	Temperature	Clivot *et al.*[22]
	T ref = 15 °C		Clivot *et al.*[22]
	PET = 584 mm	Potential evapotranspiration	Buitinik[24]
	IW = 3.35E+11 mm	Irrigation water	CBS[25]
	P = 821 mm	Precipitation	Statista[26]
	A = 79 g/kg	Clay	Hanse [27]
	$CACO_3$ = 747 g/kg	$CACO_3$ content	Clivot *et al.*[22]
	pH = 6.5	Soil pH	Hanse[27]
	CN = 9.5	C/N ratio	van der Sloot[28]
	H harvest residue = 0.315	Hummification coefficients	Clivot *et al.*[22]
	H manure = 0.525		Clivot *et al.*[22]
	H slurry = 0.15		Clivot *et al.*[22]
Bolinder *et al.*[21] model	RP = 0.766	Product allocation coefficient	Clivot *et al.*[22]
	RS = 0.219	Allocation coefficient of aboveground residue	Clivot *et al.*[22]
	RR = 0.016	Roots allocation coefficient	Clivot *et al.*[22]
	RE = 0.000	Extraroot allocation coefficient	Clivot *et al.*[22]

while the *ecoinvent* database was used for the background processes. One of the main limitations of the current study is associated with the use of a static LCI. Due to static databases, we will not be able to reflect on the dynamics of the technology with time. For example, to take into account the impact of electricity mix, a set of dynamic LCIs would be required. As we aimed to show the application of BBM-LCA through an illustrative example, we have simplified the process steps (e.g. product manufacturing in Table 2). It should be noted that the description of each part of the value chain has been detailed in the references reported in the 'modeling of greenhouse emissions associated with virgin PLA production' section.

Results and discussion

Modeling greenhouse emissions associated with the soil

Soil heterotrophic respiration (i.e., the release of CO_2 from the soil through the decomposition of plant) has been modeled using the soil carbon model of AMG described earlier. Recently, El Kadi[23] adapted the AMG model to a specific case in the Netherlands by using the local parameters, which have also been used in this paper. Table 1 provides a description of the parameters used for applying this model to sugar beet cultivation and the data from the Bolinder *et al.*[21] model.

Modeling of greenhouse emissions associated with virgin PLA production

Having calculated the carbon exchange using the AMG model presented in the previous section, we can proceed to calculate the carbon footprint associated with the first lifecycle stage – the feedstock cultivation phase. An inventory of GHG emissions occurring throughout this lifecycle stage is constructed using secondary data sourced from the scientific literature, and the *ecoinvent* database, which has been reported in Table 2.

In the feedstock cultivation phase, emissions were separated into two categories: (i) field emissions – GHGs emitted directly from the soil – and (ii) process emissions, linked to the production and procurement of agrochemicals, as well as the use of agricultural machinery for farming activities. Field and process emissions were both considered to be emitted in the first year of the lifecycle.

The lifecycle phases of material production and product manufacturing were considered downstream from feedstock cultivation. Emissions collected for these phases mainly included emissions associated with their energy consumption as well as the procurement and production of auxiliary materials and chemicals used during processing. In a similar fashion to emissions incurred by the cultivation phase, emissions occurring during material and production manufacturing were considered to be emitted in the first year of the lifecycle.

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Table 2. The source of LCI and emissions of lifecycle steps.

Phase of lifecycle	Static LCI		Emissions		Year of emissions
	Lifecycle steps	Source	GHG emissions incurred ($kgCO_2eq./kg$ PLA)	Source	
Feedstock cultivation	Process emission – agrochemicals	Garcia Gonzalez and Bjornsson[29]	4.85E–02	Garcia Gonzalez and Bjornsson[29]	Year 1
	Process emission – machinery	Garcia Gonzalez and Bjornsson[29]	3.06E–02	Garcia Gonzalez and Bjornsson[29]	
	Process emission – road transport	Garcia Gonzalez and Bjornsson[29]	1.29E–02	Garcia Gonzalez and Bjornsson[29]	
	Field emissions – NPP	The current study	–2.88E+00	The current study	
	Field emissions – heterotrophic respiration		8.22E–01		
	Field emissions – nitrous oxide		1.29E–01		
PLA production	Sugar production	Garcia Gonzalez and Bjornsson[29]	2.77E–01	Garcia Gonzalez and Bjornsson[29]	
	Lactic acid production	Morao and de Bie[30]	1.45E+00	Morao and de Bie[30]	
	Lactide production	Morao and de Bie[30]	5.24E–01	Morao and de Bie[30]	
	Polylactide production	Morao and de Bie[30]	1.67E–01	Morao and de Bie[30]	
Product manufacturing	Simplified manufacturing step	Plastic injection molding: ecoinvent[31]	8.98E–01	Kouamé[15]	
End-of-life treatment	Mechanical recycling	Cosate de Andrade *et al*.[32]	–2.11E+00	Current study	The last year of product lifetime
	Chemical recycling – depolymerization	Schwarz *et al*.[33]	–1.81E+00		
	Chemical recycling – hydrolysis	Cosate de Andrade *et al*.[32]	–8.21E–01		
	Digestion	Schwarz *et al*.[33]	–5.58E–01		
	Composting	Cosate de Andrade *et al*.[32]	5.92E–03		
	Incineration	El Kadi[23]	1.51E+00	El Kadi[23]	

Dynamic LCI

A dynamic inventory was calculated for each scenario by compiling yearly GHG emissions occurring throughout the various lifecycle stages of the considered product, resulting in the data presented in Table 3.

In year 1, the first rug was made and sugar beets were planted to renew the resource. These sugar beets are assumed to grow over the year. The carbon exchange through the soil and sugar beets, calculated by AMG model, is added to carbon footprint associated with the agricultural processes, material production, and product manufacturing, in order to provide the total CO_2 equivalent in year 1. It should be noted that the impact of plantation prior to year 1 has been included as the initial carbon stock was included in the soil carbon model.

The negative NPP in Table 3 implies that the sugar beet under the conditions in this study serves as a carbon sink even when it adds up to the heterotrophic respiration and nitrogen oxide emissions. However, to produce virgin PLA we need to go through the following steps: agrochemicals, machinery, road transport, sugar production, lactic acid production, lactide production, polylactide production as reported in Table 2, which makes the total emissions a positive number. This means that despite the fact that virgin

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Table 3. Dynamic LCI ($kgCO_2eq./kg$ PLA).

Year	Chemical recycling-depolymerization	Chemical recycling-hydrolysis	Mechanical recycling	Composting	Anaerobic digestion	Incineration
1	Virgin PLA (1.48E+00) = NPP (−2.88E+00) + heterotrophic respiration (8.22E−01) + nitrogen oxide (1.29E−01) + process emission of feedstock cultivation (9.20E−02) + PLA production (2.42E+00) + product manufacturing (8.98E−01)					
2–24	0					
25	Depolymerization (−5.07E−01) = The process itself (8.25E-01) + 95% avoided virgin PLA (−1.41E+00) + 5% virgin PLA (7.40E−02)	Hydrolysis (4.86E−01) = The process itself (1.88E+00) + 97% avoided virgin PLA (−1.44E+00) + 3% virgin PLA (4.44E−02)	Mechanical recycling (−1.36E−01) = The process itself (5.44E−01) + 96% avoided virgin PLA (−1.42E+00) + 50% virgin PLA (7.40E−01)	Composting (1.49E+00) = The process itself (6.56E−03) + avoided compost (−5.66E−04) + virgin PLA (1.48E+00)	Anaerobic digestion (9.24E−01)= The process itself (6.73E−02) + avoided electricity (−4.13E−01) + avoided heat (−2.10E−01) + virgin PLA (1.48E+00)	Incineration (2.99E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01) + virgin PLA (1.48E+00)
26–49	0					
50	Depolymerization (−5.07E−01) = The process itself (8.25E−01) + 95% avoided virgin PLA (−1.41E+00) + 5% virgin PLA (7.40E−02)	Hydrolysis (4.86E−01) = The process itself (1.88E+00) + 97% avoided virgin PLA (−1.44E+00) + 3% virgin PLA (4.44E−02)	Incineration (2.99E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01) + virgin PLA (1.48E+00)	Composting (1.49E+00) = The process itself (6.56E−03) + avoided compost (−5.66E−04) + virgin PLA (1.48E+00)	Anaerobic digestion (9.24E−01)= The process itself (6.73E−02) + avoided electricity (−4.13E-01) + avoided heat (−2.10E−01) + virgin PLA (1.48E+00)	Incineration (2.99E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01) + virgin PLA (1.48E+00)
51–74	0					
75	Depolymerization (−5.07E−01) = The process itself (8.25E−01) + 95% avoided virgin PLA (−1.41E+00) + 5% virgin PLA (7.40E−02)	Hydrolysis (4.86E-01) = The process itself (1.88E+00) + 97% avoided virgin PLA (−1.44E+00) + 3% virgin PLA (4.44E−02)	Mechanical recycling (−1.36E−01) = The process itself (5.44E−01) + 96% avoided virgin PLA (−1.42E+00) + 50% virgin PLA (7.40E−01)	Composting (1.49E+00)= The process itself (6.56E−03) + avoided compost (−5.66E-04) + virgin PLA (1.48E+00)	Anaerobic digestion (9.24E-01)= The process itself (6.73E−02) + avoided electricity (−4.13E−01) + avoided heat (−2.10E−01) + virgin PLA (1.48E+00)	Incineration (2.99E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01) + virgin PLA (1.48E+00)
76–99	0					
100	Incineration (1.51E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01)			Composting (6.00E−03)= The process itself (6.56E-03) + avoided compost (−5.66E−04)	Anaerobic digestion (−5.56E−01)= The process itself (6.73E−02) + avoided electricity (−4.13E−01) + avoided heat (−2.10E−01)	Incineration (1.51E+00)= The process itself (2.21E+00) + avoided electricity (−6.99E−01)

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material offers the benefit of additional carbon uptake, we need to explore recycling options. During the pause years, we have also assumed that we do not have any cultivation, thus resulting in no more NPP. This means that the only possibility for carbon offsetting was the cultivation years.

The EoL phases considered for the disposal of the final product were the following: chemical recycling-depolymerization, chemical recycling-hydrolysis, mechanical recycling, composting, anaerobic digestion, and incineration. In year 25, the biogenic carbon of the rug is released into the atmosphere (incineration, composting, and digestion scenarios), or recycled, prolonging the carbon storage (recycling scenarios). That same year, the second rug is either built from raw materials and sugar beets are planted again (incineration, composting, and digestion scenarios), or results from the recycling of the first rug (recycling scenarios). Regarding chemical recycling, recycled materials can be reused directly in the product lifecycle that follows. Regarding mechanical recycling, 1 kg of PLA recyclate can replace 0.5 kg of virgin PLA. In other words, it is assumed that 50% virgin PLA is required, which is based on the substitution factor suggested by Maga *et al.*[34] This is because the quality of PLA is degraded through the recycling process; to keep the quality of the recycled rug as high as the original rug, we need to add the virgin PLA.

Finally, in year 100, the fourth rug is burned (recycling and incineration scenarios). Emissions associated with these phases, including emissions credits attributed to the recovery of electricity and heat from the incineration, are considered to be emitted in the year the product is disposed of.

Instantaneous and cumulative impacts

Figure 5 illustrates the instantaneous and cumulative impacts on radiative forcing for four rugs, calculated using the dynamic LCA approach with the dynCO_2 tool.[35] The scenario analysis provides valuable insights into the relative benefits of different EoL treatments.

For the first 25 years, which is the lifetime of the first rug, the impact is the same for every scenario. In year 1, there is an increase in instantaneous radiative forcing, caused by GHG emissions related to the manufacturing of the first rug. During the following 75 years, the sequestration of carbon in the technosphere contributes to the decrease in radiative forcing in the case of depolymerization. At years 25, 50, and 75, the incineration, composting, and digestion scenarios show another increase in radiative forcing due to the manufacturing of the new rug, from the burning of the old rug (incineration scenario), and from the released GHG (composting and digesting scenarios). At year 100, the pulse of emissions due to incineration (recycling and incineration scenarios) shows a significant increase in instantaneous radiative forcing caused by the burning of the fourth rug, which is not the case for the biodegradation scenarios (composting and digestion). Both the biodegradation scenarios have their fourth rug biodegraded so that the EoL emissions are low.

The cumulative radiative forcing results indicate that the incineration scenario has the greatest impact on global warming. This is due to the absence of permanent carbon sequestration (100 years)[36] in the EoL phase, as the rugs are burned. In contrast, recycling scenarios delay the emissions of biogenic carbon, as the rugs are recycled. However, the increase in radiative forcing for the incineration scenario is lower than expected because of the avoided emissions associated with heat recovery from burning the old rug.

In the biodegradation scenarios, the increase in radiative forcing is lower than in the incineration scenario due to (i) the credit for avoided products, such as compost, and (ii) the lower carbon release as GHG. The credit is applied to compost in the composting scenario and to heat and electricity in the digestion scenario, resulting in a lower increase in radiative forcing for the digestion scenario compared with the composting scenario.

Regarding the GWI_{rel} of mechanical recycling, an inflection point is observed at 50 years because of the use of recycled PLA, which implies negative emissions is finished. In other words, the mechanical recycling scenario caused higher GHG emissions in year 50 compared with year 25 because of the use of virgin materials for the third rug, which implies additional emissions. This is due to the substitution factor suggested by Maga *et al.*[34] as 1 kg of PLA recyclate is replaced with 0.5 kg of virgin PLA.

The instantaneous impact of the depolymerization scenario continues to decrease as none of the emissions mentioned above occurs. In other words, although there are emissions from chemical recycling activities, they are lower than the emissions caused by the manufacturing of the new rug in other scenarios. Regarding the GWI_{rel} of depolymerization, an inflection point is observed at 100 years because of the incineration at year 100, which implies that positive emissions occur. The GWI_{rel} curve of depolymerization slowly tends to move away from its near-zero value, which is given by the radiative forcing caused by the residual CO_2 in the atmosphere as well as the new CO_2 from incineration. In the case of hydrolysis, the GWI_{rel} curve shows a different trend as the concentration following a net pulse emission never goes back to the pre-emission level. However, depolymerization remains near zero due to the lower impact in each cycle, giving enough space for the decay of CO_2.

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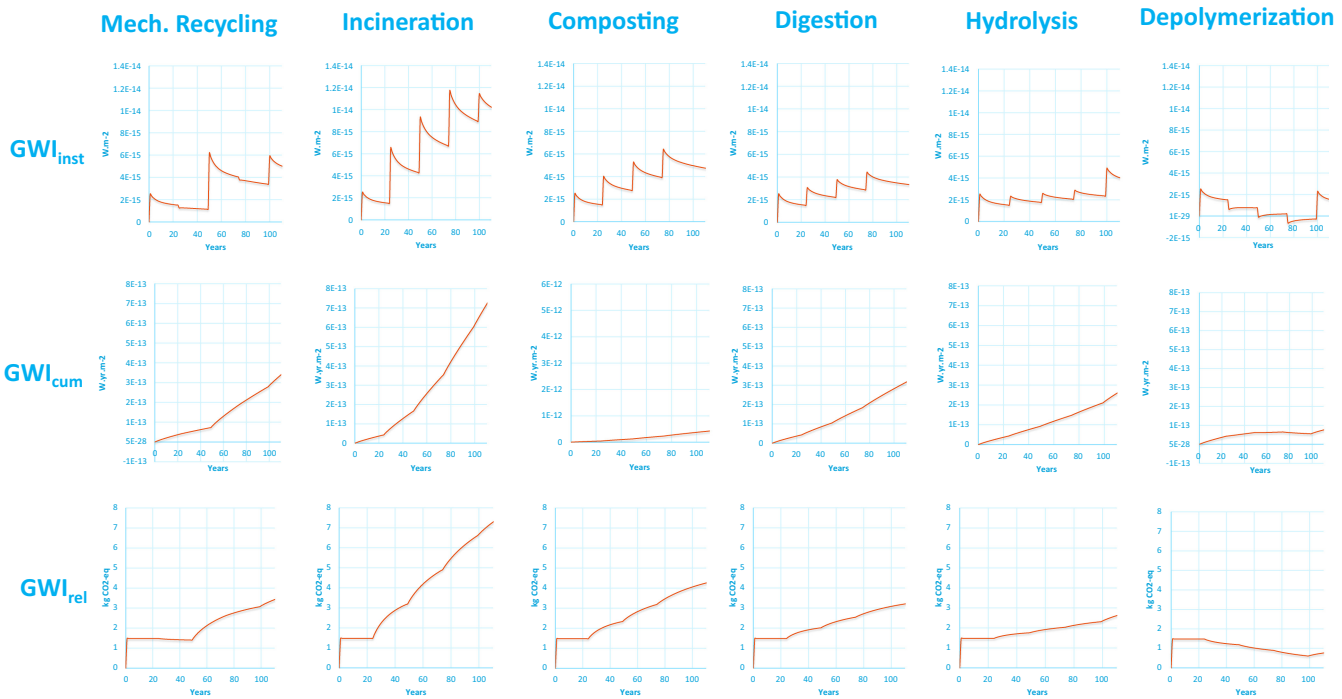


Figure 5. Instantaneous, cumulative, and relative climate change impacts.

For incineration and biodegradation scenarios, new agricultural crops are planted at years 25, 50, and 75 to replace those that have been cut to build the new rug. The carbon sequestration in the period of plantings is not big enough to contribute to the decrease in radiative forcing. The virgin PLA offers the benefit of additional carbon absorption, and a part of the sequestered carbon goes into the soil and stays there, which is considered a permanent sequestration as long as the land stays planted. However, recycling remains the most favorable treatment scenario. In other words, no inflection point due to harvesting is observed for all these scenarios although the sequestration of carbon in sugar beets implies negative emissions.

Although the recycling scenario, where recycled material from the first product is incorporated into the production of the second, came out as the best, it is not evident that this should be the case systematically. Levasseur *et al.*,[37] in a detailed analysis of wooden products, found that when energy recovery from the incineration is considered, the production of the products from virgin materials can present a better climate option than refurbishing the initial product. Nevertheless, the annual agricultural crop shows a different trend in comparison with harvested wood products with long rotation periods. As highlighted by Kouamé,[15] whereas reusing recycled material from the first product cycle saves in manufacturing of virgin material, it also forgoes the benefits of sequestering carbon by agricultural crop grown for virgin material. Although, on the one side, successively growing new agricultural crop and re-emitting the sequestered carbon back to the atmosphere at product EoL in the case of incineration without energy recovery, offers no benefit, on the other hand, when the procurement of energy from the incineration process is considered, successive remanufacturing of virgin products can have a climate advantage. The relative advantage of manufacturing a new product is in fact dependent on how the savings from the energy recovery relate to the impacts of treating the product by recycling and the impacts incurred for material production. When the recycling process generates significant GHG emissions, and the GHG savings from energy recovery are relatively high (though these savings will decrease as the share of renewable energy in electricity mixes increases), it may be more beneficial to produce two successive biobased products rather than recycling the first into the second, provided that the impacts of manufacturing new material, which are avoided in recycling, are not excessively high.

Conclusion

Biobased materials and products-life cycle assessment (BBM-LCA), as a dynamic and holistic LCA methodology, has been presented through an example in this paper. The objective of the example was not to draw a general conclusion about

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different EoL scenarios for PLA, but to show how BBM-LCA can simultaneously address both biogenic CO_2 and timing issues. The dynamic LCA enables BBM-LCA to address the common temporal issues. BBM-LCA can consider the impact of the timing of the carbon emissions and the influence of the rotation periods related to the biomass growth. More specifically, BBM-LCA can consider the potential time lag between biomass removal and regrowth. For long-lived biobased materials and products, for which GHG emissions are occurring at different moments over several years, the timeframe covered by the BBM-LCA results was also consistent with the chosen time horizon for GWPs. BBM-LCA was also able to translate the benefits of prolonging biogenic carbon storage in biobased materials and products. It considers the benefit that biobased products store carbon during lifetime.

Furthermore, BBM-LCA enables us to assess the environmental impact over multiple lifecycles in chemical and mechanical recycling, which implies time dynamics. Studies that focused on the single lifecycles have found different results from studies that considered the multiple lifecycles through BBM-LCA. For example, Cosate de Andrade *et al.*,[32] which presented only the single-loop analysis, found mechanical recycling to be the best option, followed by chemical recycling-hydrolysis and composting. This highlights the importance of time dynamics in accounting for biogenic carbon in recycling processes. In our study, when BBM-LCA was applied to model a three-product lifecycle, the scenario in which the initial product is disposed of by incineration appeared the least beneficial from a climate change perspective. The other scenarios, nevertheless, showed more subtle trends. The results showed that the strategy that involved using recycled material in the manufacturing of the second product was the most beneficial from a climate change perspective. Overall, for biobased materials, the relative benefit of recycling as opposed to incineration with energy recovery comes down to how the processes of material recycling, new material production, and EoL energy recovery are balanced in terms of their climate effect.

BBM-LCA enabled us to analyze the tradeoffs between product recycling and recoverable product incineration. One might have expected the scenario of incineration to be on par with the recycling scenario because the recycling scenario forgoes the additional carbon sequestration afforded by the cultivation of feedstock for new material. The additional amount of carbon embedded in the virgin material from newly grown agricultural crop, combined with the displaced energy from the incineration of the first product, could have amounted to a more beneficial climate effect. The result of this specific study shows that, despite the fact that virgin material offers the benefit of additional carbon uptake, recycling remains the most favorable treatment scenario except under cases of very unsustainable recycling processing and low recycled material quality.

The effectiveness of BBM-LCA stems from two key features that distinguish it from conventional assessments: its comprehensive treatment of feedstock carbon exchanges through a full ecosystem approach (i.e., NEE) and its dynamic methodology for calculating lifecycle climate impact (i.e., dynamic LCA). The PLA example demonstrates that BBM-LCA, aligned with the four steps of the ISO-14040 standards, provides a holistic and dynamic approach to estimate effectively the global warming potential of various EoL scenarios for biobased materials, offering new insights into the sustainability of biobased products.

Author contributions

Ali Ghannadzadeh was responsible for conceptualization, methodology, software development, validation, formal analysis, investigation, data curation, original draft writing, and visualization. Yvonne van der Meer acquired funding and, together with Ali Ghannadzadeh, contributed to the review and editing of the manuscript. Both authors have read and approved the final version of the manuscript.

Funding information

The financial support of the Interreg VI A Deutschland-Nederland for the project of Realise-Bio is gratefully acknowledged.

Conflict of interest statement

The authors declare no conflict of interest.

Data availability statement

The authors declare that the data supporting the findings of this study are available within the article and the cited references.

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